

MORDEN, MB, CanadaWMO: **715640**Lat: **49.1875N**Lon: **98.0839W**Elev: **976**StdP: **14.18**Time Zone: **-6.00 (NAC)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-14.5	-26.0	1.3	-18.3	-21.7	1.7	-13.9	22.6	23.0	20.6	18.4	7.0	280	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	87.1	70.6	84.2	68.6	81.2	66.5	74.1	82.7	71.6	80.4	69.3	77.8	9.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	120.3	79.1	68.6	108.8	76.8	66.1	99.9	74.6	38.4	83.3	36.1	80.6	34.0	77.9	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.1	18.0	16.2	-25.5	93.9	4.6	3.0	-28.8	96.1	-31.5	97.9	-34.1	99.6	-37.4	101.8		
(5)			-25.8	77.8	4.5	3.8	-29.0	80.5	-31.7	82.7	-34.2	84.8	-37.5	87.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	8.7	9.4	24.3	40.4	53.1	63.4	68.7	66.7	57.8	43.2	27.5	12.3
	DBStd	24.12	14.71	12.86	13.38	10.47	9.02	6.42	5.62	6.17	8.19	9.22	11.90	13.10
	HDD50	5730	1280	1137	800	315	73	2	0	0	22	255	677	1169
	HDD65	9545	1745	1557	1263	737	381	103	28	52	242	678	1125	1634
	CDD50	2001	0	0	2	28	168	404	580	518	256	43	2	0
	CDD65	342	0	0	0	0	12	55	144	105	25	1	0	0
	CDH74	3244	0	0	0	6	185	522	1226	994	287	24	0	0
	CDH80	887	0	0	0	0	46	112	333	318	73	5	0	0
(14) Wind	WSAvg	7.9	8.3	8.0	8.3	8.6	8.7	7.7	6.9	6.9	7.6	8.2	8.2	8.0
(15) Precipitation	PrecAvg	21.3	0.8	0.7	1.3	1.9	2.6	3.2	2.8	2.8	2.1	1.4	0.9	0.9
	PrecMax	30.9	2.6	2.5	4.3	5.6	8.7	7.4	5.9	7.8	6.2	4.8	2.2	1.9
	PrecMin	11.5	0.0	0.1	0.2	0.0	0.1	0.6	0.3	0.4	0.2	0.1	0.1	0.1
	PrecStd	4.8	0.6	0.6	1.0	1.5	2.0	1.6	1.6	1.9	1.4	1.2	0.6	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	42.2	61.6	74.1	86.0	87.1	90.5	92.2	86.7	77.1	60.4	43.0
		MCWB	37.6	36.5	49.7	55.1	62.0	71.7	73.8	72.6	67.9	59.7	50.1	37.3
	2%	DB	38.1	37.5	53.3	68.0	79.0	83.1	86.6	87.3	81.3	68.7	52.8	37.7
		MCWB	34.4	34.0	45.8	52.5	58.0	65.8	72.4	69.8	64.9	55.0	44.2	33.8
	5%	DB	34.1	33.0	46.0	63.3	74.1	79.8	83.8	83.4	76.8	62.6	47.8	33.6
		MCWB	31.9	30.8	40.5	49.5	56.1	64.8	70.6	68.4	62.6	51.8	41.6	31.2
	10%	DB	29.9	29.0	40.6	58.2	69.1	76.5	80.9	79.7	71.9	57.7	43.1	30.4
		MCWB	28.0	27.1	36.7	47.2	55.0	63.3	68.2	65.8	60.2	49.5	38.4	28.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	36.9	53.5	58.9	65.7	75.6	78.8	76.5	72.0	61.3	50.7	37.8
		MCDB	41.8	41.0	58.9	67.6	76.8	83.2	86.2	85.7	82.6	72.3	58.8	42.3
	2%	WB	34.6	34.1	46.2	54.0	62.3	71.1	75.1	72.7	67.7	56.8	45.5	34.0
		MCDB	37.6	37.1	53.8	65.1	70.7	78.9	82.8	82.7	75.3	66.1	51.6	36.9
	5%	WB	31.8	31.0	40.6	50.6	59.4	67.4	72.6	70.2	64.8	53.0	41.9	31.5
		MCDB	34.2	33.1	45.4	60.9	68.4	75.5	80.7	79.6	73.1	61.0	47.1	33.5
	10%	WB	28.0	27.1	37.0	47.9	56.8	64.8	70.1	67.8	62.0	49.9	38.7	28.7
		MCDB	29.8	29.1	40.3	57.0	66.7	73.3	78.6	76.2	70.3	57.2	42.9	30.4
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	15.7	15.7	19.5	20.8	18.8	19.3	20.4	20.1	16.5	14.3	13.8
		MCDBR	18.3	18.0	21.5	28.3	28.8	24.1	22.9	25.8	27.4	25.7	21.1	17.8
		MCWBR	15.9	15.1	15.2	15.9	13.9	11.9	12.2	12.2	13.9	14.2	14.3	15.3
		MCDBR	17.8	17.5	20.6	26.9	22.0	20.8	21.4	22.6	23.7	23.9	20.0	16.9
	5% WB	MCWBR	15.5	14.9	15.1	16.2	13.1	12.4	13.1	12.6	13.8	14.0	14.1	14.6
(40) Clear-Sky Solar Irradiance	taub		0.275	0.292	0.301	0.359	0.375	0.381	0.387	0.384	0.349	0.303	0.271	0.253
	taud		2.258	2.230	2.386	2.322	2.340	2.352	2.344	2.343	2.432	2.535	2.483	2.388
	Ebn at Noon		246	270	291	281	281	279	275	271	270	268	254	245
	Edh at Noon		23	30	31	37	38	38	37	36	30	23	18	18
(44) All-Sky Solar Radiation	RadAvg		473	853	1247	1577	1754	1888	2005	1648	1204	735	418	332
	RadStd		43	66	106	121	131	134	86	108	86	97	57	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	-3.35	-4.14	N/A	N/A	N/A	N/A
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.21	N/A	N/A	N/A	N/A